

Jiahua Zhu

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Education

National Institute of Natural Hazards, MEMC, CAS

Beijing, China

MASTER OF SCIENCE IN GEOLOGY

Aug. 2023 - Present

- Core Courses: Deep Learning, Data Mining, Geo-processing with Python, Remote Sensing Science

China University of Geosciences, Beijing

Beijing, China

BACHELOR IN GEOGRAPHIC INFORMATION SCIENCE, SCHOOL OF ARTIFICIAL INTELLIGENCE

Aug. 2019 - Jun. 2023

- Core Courses: Computer Language Programming, Data Structures, Principles of Spatial Database, Digital Image Processing, Principles of GIS, GIS Spatial Analysis, Image processing of Remote sensing

Publications

[P1] SCPDNet: A Network for Landslide Detection Combining Deformable Attention and Multiscale Fusion in Remote Sensing Imagery.

Jiahua Zhu, Wenliang Jiang, Qiang Li, Xue Li

IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, 2025 [\[Paper\]](#)

[P2] Amplitude-Phase Spectral Attention With Self-Calibrating Spatial-Frequency Fusion for Robust Remote Sensing Change Detection.

Jiahua Zhu, Wenliang Jiang, Qiang Li, Xue Li

IEEE Geoscience and Remote Sensing Letters, 2025 [\[Paper\]](#)

[C1] Deep Learning Methods for Cross-domain Pseudo-change Detection.

Jiahua Zhu.

Symposium on Satellite-based Earthquake Observation Technology and Applications, 2025 [\[Oral\]](#)

Research Experience

National Institute of Natural Hazards

RESEARCH ASSISTANT

Aug. 2024 - Present

- Participated in two projects: Disaster situational awareness for mining rescue robots based on deep learning, and remote sensing algorithm development, providing actionable technical analysis and decision support for government agencies.
- Translated technical workflows and research findings into deployable solutions, coordinating cross-team collaboration to facilitate cloud platform deployment and real-world application of research outputs.
- Prepared technical reports and quantitative evaluation analyses, supporting principal investigators and relevant authorities in post-disaster assessment and emergency response decision-making.

FDEF-Net - Frequency-Domain Enhanced Network for Change Detection

Apr. 2025 - Sep. 2025

- Method Innovation: Proposed FDEF-Net, utilizing amplitude-phase decoupling to enhance change detection accuracy, facilitating reliable post-disaster monitoring and rapid emergency response.
- Experimental Results: Achieved an average improvement of 1.5-2% in F1 and IoU compared with existing state-of-the-art (SOTA) models on LEVIR-CD, WHU-CD, and GZ-CD benchmarks.
- Model Efficiency: Reached competitive performance with only 4.52M parameters and 4.44G FLOPs, demonstrating a lightweight yet effective design.

SCPDNet: Deformable Attention Network for Fine-Grained Landslide Detection

Jul. 2024 - May. 2025

- Project Overview: Developed a semantic segmentation model for landslide detection based on SAM with a ViT backbone, integrating deformable convolution and cross-pyramid attention to improve boundary recognition in complex terrains.
- Main Responsibilities: Redesigned the Decoder with cross-scale deformable attention and multi-scale pooling to fuse semantic and detail features, achieving an mIoU of 87.51% with nearly 10% improvement over mainstream baselines on multiple remote sensing datasets.

KI-LoRA + Knowledge Graph Enhanced RAG for Mining Disaster LLMs

Jan. 2025 – Present

- Project Overview: Proposed KI-LoRA, a knowledge-injected fine-tuning paradigm that integrates domain knowledge into LLM training via a knowledge graph and Retrieval-Augmented Generation (RAG), aiming to mitigate hallucinations.
- Main Responsibilities: Constructed a domain knowledge graph and RAG pipeline, fine-tuned the Qwen3 model using the Unsloth framework, and evaluated with EvalScope. Demonstrated improved factual consistency and causal reasoning in domain-specific QA and emergency inference tasks compared with baseline models.

Ecological Environment Assessment of Xiong'an New Area Using RSEI on GEE

Nov. 2022 – May. 2023

- Project Overview: Evaluated the ecological environment of Xiong'an New Area over a 10-year period (2015–2023) using the Remote Sensing Ecological Index (RSEI) on the Google Earth Engine (GEE) platform.
- Main Responsibilities: Processed 70 cloud-free Sentinel-2 images and integrated four ecological indicators (NDVI, LST, WET, NDBSI). Conducted temporal-spatial trend analysis, showing an 8% improvement in ecological quality after 2017, with a 2% decline in 2023 caused by urban expansion offsetting vegetation growth.

Honors and Awards (Selected)

2024	Outstanding Student (Top 10%) , University of Chinese Academy of Sciences	Beijing, China
2023	Second Class Scholarship (Top 10%) , China University of Geosciences, Beijing (CUGB)	Beijing, China
2022	Finalist Winner (Top 1% Worldwide) , Mathematical Contest in Modeling (MCM) - Team Leader	U.S.A
2021	Meritorious Winner (Top 7% Worldwide) , Mathematical Contest in Modeling (MCM) - Team Leader	U.S.A
2021	Second Prize , National College Mathematical Modeling Competition - Team Leader	Beijing, China
2021	Second Prize , National Digital Art Design Competition, Beijing Region	Beijing, China

Academic Services

Reviewer	IEEE Transactions on Geoscience and Remote Sensing, Artificial Intelligence in Geosciences, etc.
Teaching Assistant	University of Chinese Academy of Sciences, Remote Sensing Applications for Natural Hazards (2024)
Teaching Assistant	China University of Geosciences, Surveying (2021)

Skills

Languages	Mandarin (Native), English (IELTS: 6.5)
Programming	Python, C++, Java, MATLAB, SQL, $\LaTeX$
Software	PyTorch, Git, Anaconda, ArcGIS, GEE, ENVI, QGIS